

Rajalakshmi Engineering College

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2024_28_III_OOPS Using Java Lab

2028_REC_OOPS using Java_Week 3_Q4

Attempt : 1
Total Mark : 10
Marks Obtained : 10

Section 1 : Coding

1. Problem Statement

Sesha is developing a weather monitoring system for a region with multiple weather stations. Each weather station collects temperature data hourly and stores it in a 2D array.

Write a program that can add the temperature data from two different weather stations to create a combined temperature record for the region.

Input Format

The first line of input consists of two space-separated integers N and M, representing the number of rows and columns of the matrices, respectively.

The next N lines consist of M space-separated integers, representing the values of the first matrix.

The following N lines consist of M space-separated integers, representing the values of the second matrix.

Output Format

The output prints the addition of the two matrices in N rows and M columns, representing the combined temperature record.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 3 3

1 2 3

4 5 6

7 8 9

1 1 1

2 2 2

3 3 3

Output: 2 3 4

6 7 8

10 11 12

Answer

```
// You are using Java
import java.util.Scanner;
class WeatherStationMatrixAdder {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int m = sc.nextInt();
        sc.nextLine(); // Consume newline

        int[][] station1 = new int[n][m];
        int[][] station2 = new int[n][m];

        // Read first station data
        for(int i=0; i<n; i++) {
            String[] row = sc.nextLine().split(" ");
            for(int j=0; j<m; j++) {
                station1[i][j] = Integer.parseInt(row[j]);
            }
        }
    }
}
```

```

    }
}

// Read second station data
for(int i=0; i<n; i++) {
    String[] row = sc.nextLine().split(" ");
    for(int j=0; j<m; j++) {
        station2[i][j] = Integer.parseInt(row[j]);
    }
}

// Calculate and print combined data
for(int i=0; i<n; i++) {
    for(int j=0; j<m; j++) {
        System.out.print((station1[i][j] + station2[i][j])
            + (j == m-1 ? "\n" : " "));
    }
}
}
}
}

```

Status : Correct

Marks : 10/10